

Microbiome AD Generalization

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Cross-Cohort Generalization Failure in Gut Microbiome-Based Alzheimer's Disease Classifiers: Evidence from Five Independent Cohorts with Batch Correction Analysis

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Abstract

Gut microbiome dysbiosis has been proposed as a biomarker for Alzheimer's disease (AD) and mild cognitive impairment (MCI), yet individual studies consistently report contradictory taxonomic signatures across populations, and cross-cohort classifier performance has received limited systematic evaluation. We assembled five publicly available 16S rRNA gene sequencing cohorts (509 samples; China, Kazakhstan, South Korea; Zhuang 2018, Ling 2020, Zhu 2022, Kaiyrylykzy 2022, Kim 2022) and evaluated whether machine learning classifiers trained within one cohort could generalize to independent cohorts, whether standard batch correction methods could recover transferability, and which taxonomic features drove cohort-specific predictions. DADA2 amplicon sequence variant (ASV) processing and genus-level harmonization produced a unified matrix of 396 genera across all cohorts (retained at $\geq 20\%$ prevalence in

at least one cohort, applied globally across all five cohorts before any supervised split), transformed to centered log-ratio (CLR) coordinates for compositional analysis; a dedicated sensitivity analysis re-deriving this feature set independently within each training split (never using held-out-cohort information) reproduced the same qualitative generalization-failure pattern (LOCO AUC changed by at most 0.066). The Zhu 2022 cohort deposited paired blood microbiome and fecal samples; only fecal samples (n=90) were retained for all analyses. Within four labeled cohorts (n=401 for binary AD/CN analyses), logistic regression and LightGBM classifiers achieved within-cohort AUC-ROC values ranging from 0.633 to 0.998 under nested cross-validation, broadly consistent with originally reported performance. In leave-one-cohort-out (LOCO) transfer experiments, LOCO AUC-ROC fell to 0.504–0.813, a mean drop of 0.176 AUC points for logistic regression and 0.146 for LightGBM, with one of eight model-cohort combinations collapsing to near chance (AUC = 0.504). PERMANOVA on Aitchison distance showed that cohort identity explained 19.3% of total Aitchison variance across all five cohorts ($F=30.13$, $p=0.0001$); in the four-cohort marginal model, cohort explained $R^2=0.172$ versus only 0.014 for disease label—an approximately 12-fold disparity in cohort-associated versus diagnosis-associated compositional heterogeneity, which provides context for the observed difficulty of harmonization but does not by itself establish that the two sources of variation overlap or explain why batch correction changed classifier performance. Batch correction was evaluated under two designs: a label-blind design (ComBat-seq/MMUPHin fit without any diagnosis information) left mean LOCO AUC essentially unchanged relative to uncorrected (within 0.007 AUC for both methods and models), while an exploratory label-informed design (fit using each sample’s own true diagnosis label, which cannot estimate prospective performance) produced large, method- and cohort-specific degradation, including a collapse to AUC=0.308 for one cohort under ComBat-seq — a result attributable in large part to the label-informed design itself rather than to batch correction per se. Neither design recovered cross-cohort transferability. SHAP analysis using out-of-fold values revealed near-zero pairwise Jaccard similarity between cohorts’ top-20 predictive taxa (mean 0.135 for logistic regression and 0.058 for LightGBM); against a null distribution matched to the observed statistic’s own construction (four random top-20 sets per replicate, averaging the six pairwise Jaccards among them, 100,000 replicates), both were significantly above chance at the primary $N=20$ threshold (logistic regression $p<0.0001$; LightGBM $p=0.0048$), though the LightGBM result was not significant at $N=10$ ($p=0.45$), indicating overlap that is real but small in magnitude and sensitive to the choice of N . Using a pre-specified, descriptive cross-fold coefficient-sign-stability screen for logistic regression (positive or negative sign in ≥ 8 of 10 outer folds, not a formal hypothesis test), eight genera met the criterion for opposite, stable fitted-coefficient directions across cohorts; because LightGBM is a tree ensemble with no single coefficient-like direction, no comparable screen is applied to it. These results demonstrate that gut microbiome-based AD classifiers trained in one population do not transfer reliably to other populations, that the cohort effect is structurally much larger than the disease effect, and that the batch correction methods evaluated here do not recover this transferability. Multi-cohort harmonized training data and prospective validation across geographically and demographically diverse populations are prerequisites before microbiome-based AD classifiers can be considered clinically actionable.

1. Introduction

Alzheimer's disease (AD) is the leading cause of dementia worldwide, affecting an estimated 55 million people, with projections exceeding 150 million by 2050 (World Health Organization, 2023). The gut–brain axis has emerged as a mechanistic pathway linking intestinal microbial communities to neurodegeneration: altered gut microbiome composition has been associated with AD pathology through neuroinflammatory, metabolic, and neuroendocrine routes (Cryan et al., 2019; Kowalski and Mulak, 2019). Since the landmark observation that germ-free APPPS1 transgenic mice showed reduced amyloid- β pathology in the absence of gut microbiota compared with conventionally colonized APPPS1 controls (Harach et al., 2017), dozens of 16S rRNA and shotgun metagenomic studies have reported significant differences in gut microbial composition between AD or MCI patients and cognitively normal (CN) controls. MCI is included as a diagnostic category in this literature given its role as a preclinical or prodromal stage of AD, though the primary supervised analyses in this study focus on binary AD versus CN comparisons (see Section 2.4).

The clinical appeal of a gut microbiome-based diagnostic signature is considerable: stool is non-invasive, inexpensive to collect, and microbiome profiling is increasingly accessible. Multiple independent research groups have trained machine learning classifiers on gut microbiome abundance profiles and reported promising within-cohort discrimination performance, with area under the receiver operating characteristic curve (AUC-ROC) values of 0.70–0.90 in several published studies (Zhuang et al., 2018; Ling et al., 2021; Zhu et al., 2022). However, a critical property of any diagnostic biomarker—transferability across independent populations—has rarely been tested in this literature.

The limited cross-cohort comparisons that do exist reveal a striking inconsistency in the direction of reported associations. A systematic review and meta-analysis synthesizing 17 published studies (679 AD/MCI patients, 632 controls) documented that *Bacteroides* abundance was significantly elevated in North American cohorts yet significantly reduced in Chinese cohorts—a directional reversal across geographically distinct populations (Jemimah et al., 2023). Similar contradictions have been observed for *Akkermansia*, *Faecalibacterium*, and *Lactobacillus* across individual studies (Vogt et al., 2017; Liu et al., 2019; Zhuang et al., 2018). These population-level contradictions suggest that published within-cohort classifiers may be detecting cohort-specific compositional signatures—driven by diet, geography, host genetics, sequencing protocols, or a combination thereof—rather than a universal microbial correlate of AD pathology.

A recent multi-cohort study integrating 15 public AD microbiome projects identified candidate gut microbiota indicators across multiple geographic populations using integrated data (Wang et al., 2026), confirming that multi-population integration remains a methodological frontier. The present study is complementary, providing a systematic quantification of cross-cohort generalization failure and an empirical analysis of why batch correction methods fail to bridge the gap—questions that remain open even where multi-cohort classifiers have been attempted.

Despite this recognized heterogeneity, few studies have jointly quantified: (1) the magnitude of cross-cohort AUC collapse when trained classifiers are applied to independent cohorts; (2) the relative contribution of technical batch effects versus biological heterogeneity to this collapse, as estimated by variance partitioning; and (3) whether state-of-the-art microbiome batch correction methods can recover cross-cohort generalization. Answering these three questions jointly in one rigorous analysis is the primary contribution of this paper.

We assembled five publicly available 16S rRNA gene sequencing cohorts spanning three countries (China, Kazakhstan, South Korea) and conducted a comprehensive cross-cohort generalization evaluation. We

report that: classifiers trained within one cohort consistently fail to generalize to other cohorts; cohort identity explains approximately 12 times more compositional variance than disease status; standard batch correction methods (ComBat-seq and MMUPHin) reduced mean cross-cohort AUC rather than improving transferability; and the specific taxa driving predictions are almost entirely cohort-specific and frequently contradict in direction across populations. Together, these findings constitute a systematic quantification of the generalization failure in the gut microbiome-based AD classifier literature, with direct implications for the field's trajectory toward clinical translation.

2. Materials and Methods

2.1 Cohort Identification and Data Acquisition

We performed a systematic search of NCBI SRA and ENA for publicly available 16S rRNA amplicon sequencing studies of gut microbiota in AD, MCI, or cognitively normal populations, deposited before January 2025. The search used the following terms in the SRA search interface: (“Alzheimer’s disease” OR “Alzheimer disease” OR “mild cognitive impairment” OR “MCI” OR “dementia”) AND (“gut microbiome” OR “gut microbiota” OR “intestinal microbiome” OR “16S rRNA”) AND (“Homo sapiens”). Studies were included if they provided: (1) per-sample diagnosis labels (AD, MCI, or CN) accessible in public metadata; (2) raw FASTQ files or quality-filtered reads deposited in a public archive; (3) minimum sample sizes of 40 participants per cohort. We restricted this analysis to 16S rRNA studies; shotgun metagenomic cohorts were identified but excluded to avoid the additional platform confound introduced by mixing amplicon and whole-genome sequencing data. Cohorts were excluded if: per-sample labels were IRB-restricted (for supervised analyses); data had not been publicly released at the time of analysis; or samples represented a tissue compartment other than fecal microbiome as the primary analyte. A complete record of the search procedure, identified cohorts, and exclusion rationale is provided in Supplementary Note S1.

Four cohorts met all criteria for supervised analysis. A fifth cohort, Kim/KBASE 2022 (PRJEB50447), lacked publicly available per-sample diagnosis labels and was retained only for unsupervised cohort-level analyses. The ALBION 2025 Greek cohort (PRJNA1297934) was identified but excluded because data had not yet been publicly released at the time of analysis. Group-level counts from the published paper (18 amyloid-positive preclinical AD; 60 cognitively normal) were available but insufficient for supervised classifier training or evaluation.

The four cohorts with per-sample diagnosis labels used in supervised analyses were:

Zhuang 2018 (China; PRJNA554111): 86 participants (43 AD, 43 CN) from the Third Military Medical University, Chongqing. Paired-end MiSeq 2×300 bp, V3–V4 region (Zhuang et al., 2018).

Ling 2020 (China; PRJNA633959): 171 participants (100 AD, 71 CN). Paired-end MiSeq 2×300 bp, V3–V4 region (Ling et al., 2021).

Zhu 2022 (China; PRJNA489760): 180 samples deposited in SRA from a larger 302-participant study, representing 90 participants (30 AD, 30 MCI, 30 CN) each contributing 2 sample types = 180 samples. Each participant contributed two samples: a fecal (F_) *sample and a blood microbiome (B_) sample*. Only the 90 fecal samples (30 AD, 30 MCI, 30 CN) were used in all analyses; the 90 blood microbiome samples were excluded as they represent a distinct biological compartment and would confound within-cohort

classifiers. Binary analyses used 30 AD and 30 CN fecal samples (n=60). Paired-end MiSeq 2×300 bp, V3–V4 (Zhu et al., 2022). Diagnostic criteria across cohorts relied on clinical AD diagnosis (Zhuang, Ling, Zhu) or amyloid-PET positivity (Kim); these differences in diagnostic ascertainment are acknowledged as a source of between-cohort biological heterogeneity.

Kazakhstan 2022 (PRJNA811324): 84 participants (41 AD, 43 CN). Single-end NovaSeq sequencing, V3–V4 region (Kaiyrlykyzy et al., 2022). This cohort is the only single-end dataset in the study; implications for read length and taxonomic resolution are discussed in Section 2.2 and the Limitations section.

The Kim/KBASE 2022 cohort (Jung et al., 2022; PRJEB50447; 78 participants; 18 preclinical AD by amyloid-PET, 60 CN) was included in PERMANOVA analyses (Phase 4) only and is explicitly excluded from all supervised classifier training and evaluation.

All raw FASTQ data were downloaded from NCBI SRA using fasterq-dump (SRA Toolkit v3.0.10). Cohort-level sequencing statistics are reported in Supplementary Table S2.

2.2 DADA2 Amplicon Sequence Variant Processing

Raw paired-end reads (four cohorts) and single-end reads (Kazakhstan) were processed independently per cohort using DADA2 v1.28 in R v4.3 (Callahan et al., 2016), to avoid introducing cross-cohort data leakage during error model estimation.

For paired-end cohorts, reads were trimmed to truncation lengths selected per cohort based on per-base quality score profiles (generally 240 bp forward / 200 bp reverse for V3–V4 MiSeq 2×300 data), filtered to remove reads with >2 expected errors, chimeras were removed using the consensus method, and forward and reverse reads were merged. For Kazakhstan (single-end), the single-end DADA2 workflow was applied without merging; truncation length was adjusted to 250 bp after inspection of quality profiles (see Supplementary Note S2 for rationale). The mixed single-end/paired-end design is an additional technical source of between-cohort heterogeneity acknowledged as a study limitation.

Taxonomic assignment used the SILVA reference database v138.1. ASVs were collapsed to genus level by summing raw counts per sample. A genus was retained across the entire study if it reached $\geq 20\%$ prevalence (i.e., a nonzero count in $\geq 20\%$ of samples) in at least one of the five cohorts; this prevalence threshold was computed per cohort and the union of genera passing in any cohort was taken across all five cohorts, producing a unified feature set of 396 genera. Genera absent in a given sample received a count of zero. Because this feature-selection step used all five cohorts' raw prevalence simultaneously, the resulting 396-genus feature universe was informed, in part, by the prevalence of the eventual test cohort in every downstream leave-one-cohort-out (LOCO) and pairwise-transfer experiment (Section 2.5) — a global feature-selection design whose held-out-cohort information use we evaluate directly in Section 2.5.1.

2.3 Compositional Data Transformation

Raw genus-level count tables were converted to centered log-ratio (CLR) coordinates for all downstream analyses requiring distributional assumptions on unconstrained data (machine learning classifiers, PERMANOVA on Euclidean distance). The CLR transformation is defined as:

$$\text{clr}(x_i) = \log \left(\frac{x_i}{g(x)} \right)$$

where $g(x)$ is the geometric mean of the composition x .

Zero counts were replaced prior to transformation using the multiplicative replacement strategy (Martín-Fernández et al., 2003) with $\delta = 0.65$:

$$x_i^* = \begin{cases} \delta & \text{if } x_i = 0 \\ (1 - n_0 \cdot \delta / T) \cdot x_i & \text{if } x_i > 0 \end{cases}$$

where $T = \sum_j x_j$ is the total read count of the sample and n_0 is the count of zeros in the sample. The unified genus matrix contained 75.4% zero entries (151,926 of 201,564 cells across the 509 fecal samples and 396 genera). The value $\delta = 0.65$ was chosen such that the replacement mass remained small relative to typical per-sample sequencing depth (median 29,521 reads): in the worst-observed sample, the total mass allocated to zero-filled entries was 2.53% of that sample's total reads. A built-in safeguard rescales δ downward whenever $n_0 \cdot \delta \geq T$, capping replacement mass at 1% of sample total in such cases. This strategy preserves the ratios among non-zero components while introducing a minimal perturbation proportional to the relative frequency of zeros, and is preferable to the commonly used additive pseudocount which distorts low-abundance taxa more severely.

CLR-transformed values were used as features for all machine learning models and for Aitchison distance computation (Euclidean distance on CLR values). Raw genus-level count matrices (not CLR-transformed) were provided as input to ComBat-seq, which requires non-negative integer counts. Relative abundances (proportions) were provided to MMUPHin. Bray-Curtis dissimilarity was computed on relative abundances. CLR values were used for all other analyses.

2.4 Within-Cohort Classifier Baselines

We trained binary classifiers (AD vs. CN) independently for each of the four labeled cohorts using two model families:

L2-regularized logistic regression (scikit-learn v1.5; Pedregosa et al., 2011; LogisticRegression, solver="saga"). Regularization strength C was tuned over $\{0.001, 0.01, 0.1, 1.0, 10.0\}$ in inner cross-validation. An `l1_ratio` parameter was set in the estimator configuration but has no effect under the default L2 penalty actually used (scikit-learn's `penalty` argument was not set to "elasticnet"); all logistic-regression results in this paper are therefore L2 (ridge)-regularized, not elastic-net, despite the model being informally referred to as "elastic-net" in earlier drafts of this work. This is corrected here for accuracy; it does not require re-fitting any model, since the actual fitted models were always L2-regularized.

LightGBM gradient-boosted trees (lightgbm v4; Ke et al., 2017). Hyperparameters tuned: number of estimators $\in \{50, 100, 200\}$, learning rate $\in \{0.05, 0.1\}$, max depth $\in \{3, 5\}$, number of leaves $\in \{15, 31\}$.

For Zhu 2022, AD was defined as the positive class and MCI samples were excluded from binary analyses (resulting in $n=60$: 30 AD, 30 CN fecal samples); a separate three-class analysis (AD / MCI / CN) used all 90 fecal samples and is reported as a secondary finding. Because only fecal samples were retained, each patient appears exactly once; standard StratifiedKFold was used without any paired-sample grouping structure.

Performance was estimated using nested cross-validation: a 10-fold outer loop for performance estimation and a 5-fold inner loop for hyperparameter tuning. The primary metric was AUC-ROC computed on out-of-fold predictions pooled across the 10 outer folds, providing a single unbiased AUC estimate per cohort-model combination. Bootstrap 95% confidence intervals (1,000 resamples) on the pooled out-of-fold predictions are reported throughout. We also report sensitivity and specificity at the Youden’s J optimal threshold. For comparison with originally reported performance, the published AUC values from the original cohort papers are noted in Table 2 where available.

2.5 Cross-Cohort Generalization Experiment

We evaluated two cross-cohort transfer designs:

Leave-one-cohort-out (LOCO): For each test cohort, a classifier was trained on the pooled samples from the remaining three labeled cohorts (no cross-validation within this training step; hyperparameters were selected via 5-fold inner CV on the training pool). The trained model was then evaluated on the held-out cohort. This produced four LOCO AUC estimates per model type. We report the LOCO AUC drop as the difference between within-cohort (nested CV) AUC and LOCO AUC; bootstrap 95% CIs are reported for each LOCO AUC estimate, and the non-overlap of within-cohort and LOCO CIs serves as an indicator of a statistically reliable drop.

Pairwise transfer: A classifier was trained on one cohort and tested on each of the other three cohorts individually, yielding a 4×4 matrix (diagonal excluded) of cross-cohort AUC values per model type. This design is more stringent than LOCO and isolates specific directional failures.

Bootstrap 95% CIs (1,000 resamples) on the test set predictions are reported for all LOCO and pairwise estimates.

Sensitivity analyses: To assess whether generalization failure was attributable to any single cohort, we repeated the LOCO experiment under three additional configurations: (A) excluding Kazakhstan entirely, with LOCO evaluated on the remaining three cohorts; (B) excluding Zhuang 2018; (C) excluding Ling 2020. Results are reported in Section 3.3 and Supplementary Figure S4.

2.5.1 Training-Only Feature-Selection Sensitivity Analysis

Because the 396-genus feature set (Section 2.2) was selected globally using all five cohorts’ raw prevalence, every LOCO and pairwise experiment above used a feature universe partly informed by the eventual held-out cohort. To assess whether this affected the generalization-failure conclusion, we repeated the within-cohort nested-CV, LOCO, and pairwise-transfer analyses with genus selection re-derived independently inside every individual split, using only that split’s training-side samples: for each outer fold (within-cohort), each held-out cohort (LOCO), and each train cohort (pairwise), the $\geq 20\%$ -prevalence filter was recomputed from scratch using only the training-side AD/CN-labeled samples for that specific split, and the corresponding test/held-out samples were projected onto the training-derived genus set (absent genera filled with zero count) before per-sample zero-replacement and CLR transformation. Model families, hyperparameter grids, random state, fold counts, and the bootstrap CI procedure were identical to the main analyses (Sections 2.4–2.5). This produces a strictly train-only feature universe for every split, at the cost of a smaller and split-specific genus count (e.g., 150 genera when Kazakhstan is excluded from feature selection during its own LOCO hold-out, versus 396 in the main global-feature analysis).

2.6 PERMANOVA Variance Decomposition

To quantify the relative contribution of cohort identity and disease status to total compositional variance, we conducted PERMANOVA (permutational multivariate analysis of variance) using the `adonis2` function in the R package `vegan` v2.6 (Oksanen et al., 2024). Two distance matrices were used:

- **Aitchison distance** (Euclidean distance on CLR values): the compositionally appropriate distance that does not distort due to the simplex constraint.
- **Bray-Curtis dissimilarity** (on relative abundances): included for comparability with the broader microbiome literature.

All PERMANOVA models used 9,999 permutations with Type III (marginal) sums of squares (by="margin") to allow independent assessment of cohort and diagnosis effects. The minimum attainable permutation p-value with 9,999 permutations is reported as $p=0.0001$ throughout. Three models were fitted: (1) cohort only, all five cohorts ($n=509$); (2) cohort + diagnosis, restricted to the four labeled cohorts ($n=401$: AD and CN samples only, Zhu 2022 MCI excluded), with marginal R^2 for each term; (3) within each labeled cohort, diagnosis as the sole predictor (within-cohort disease effect size). Note that Figure 4B displays the PCoA from Model 1 ($n=509$, all five cohorts including MCI samples), while the marginal R^2 values reported throughout the text derive from Model 2 ($n=401$).

Homogeneity of multivariate dispersions was tested using `betadisper` and `permutest` (9,999 permutations) to assess whether observed cohort differences reflected centroid separation, within-group spread heterogeneity, or both. A significant beta-dispersion test indicates that PERMANOVA R^2 values partially reflect variation in within-group spread, not only centroid shifts.

To further quantify the degree to which cohort-specific information is retained in the preprocessed feature space, we trained a logistic regression classifier (one-vs-rest, stratified 5-fold CV) to predict cohort identity from the CLR-transformed features across all 509 samples. The macro-AUC for cohort prediction provides a complementary measure of cohort separability.

2.7 Batch Correction

We evaluated one count-based batch-correction method (ComBat-seq) and one microbiome-specific meta-analysis framework (MMUPHin), applying each with cohort identity as the batch variable, under two distinct designs that differ in whether diagnosis (disease-label) information is supplied during correction. Both designs are **transductive**: both methods estimate batch-effect parameters jointly across every sample being corrected — including, in the downstream LOCO evaluation, the eventual held-out test cohort's own feature values and cohort membership. Neither design is prospective or inductive (i.e., neither defines a transformation that could be applied to a genuinely new, unseen patient without refitting); they differ only in whether the target cohort's diagnosis labels are additionally used.

Design 1 — label-blind transductive correction (primary). ComBat-seq (Zhang et al., 2020) was run on raw genus-level count matrices (genera $\times$ samples, four labeled cohorts only) with `group = NULL`, `full_mod = FALSE` — no diagnosis information of any kind is supplied. MMUPHin v1.15 (Ma et al., 2022) was run on relative-abundance profiles from the four labeled cohorts with `batch = cohort` and `covariates = NULL`; the metadata object supplied to `adjust_batch` did not include a diagnosis column at all. In both cases, cohort identity and every sample's own raw feature values are used (this is what makes the design transductive), but no sample's true AD/CN label is ever used during correction.

Corrected count tables (ComBat-seq) or relative abundances (MMUPHin) were CLR-transformed using the same multiplicative zero-replacement procedure described in Section 2.3.

Design 2 — label-informed transductive correction (exploratory/oracle-style sensitivity). The same two methods were additionally run with `group = disease_status`, `full_mod = TRUE` (ComBat-seq) and `covariates = diagnosis` (MMUPHin) — i.e., using every sample’s own true diagnosis label, for all four labeled cohorts jointly, including the sample’s own held-out-cohort membership in the eventual LOCO evaluation. This design cannot estimate prospective external-validation performance, because the correction step has already seen the answer for every sample it corrects, including test-cohort samples. We retain it only as an exploratory, oracle-style sensitivity analysis illustrating an upper/alternative bound on what correction with label information could do, not as evidence about deployable performance.

After each correction, the LOCO generalization experiment (Section 2.5) was repeated on the corresponding batch-corrected CLR matrices, and within-cohort AUC on corrected data was re-computed under the same nested-CV procedure as Section 2.4. Because Design 2’s correction step is explicitly told each sample’s own diagnosis label as part of a global, jointly-fit model, a within-cohort AUC measured on Design-2-corrected features cannot be used to argue that “genuine” disease-associated biological signal survived correction independent of that label information — some or all of any preserved within-cohort separability could instead be an artifact of the correction algorithm’s own label-preservation mechanism (`full_mod=TRUE / covariates="diagnosis"`) rather than evidence about biology. This caveat does not apply to Design 1, whose within-cohort AUC on corrected data is not confounded by label use during correction, though it remains a transductive (not prospective) result like the LOCO evaluation itself.

2.8 SHAP Feature Importance, Direction, and Cross-Cohort Overlap Analysis

To identify which genera drove predictions in each cohort and whether these features were consistent across cohorts, we computed SHapley Additive exPlanations (SHAP) values (Lundberg and Lee, 2017) using an out-of-fold (OOF) procedure to avoid the in-sample overfitting bias that arises when SHAP values are computed on the same data used to fit the model.

For each labeled cohort, SHAP values were computed using the same 10-fold outer CV structure used for AUC estimation (Section 2.4). In each fold, the classifier was trained on the 9 training folds; SHAP values were then computed for the held-out fold using the training fold as the SHAP background distribution. For logistic regression, `shap.LinearExplainer` was used with the training-fold data as background (`feature_perturbation="interventional"`), and the held-out fold as the evaluation set; this ensures the mean SHAP value over the evaluation set is not algebraically forced to zero (as would occur if background and evaluation sets were identical). For LightGBM, `shap.TreeExplainer` was used (`check_additivity=False`) with the same OOF structure. SHAP values were aggregated across all 10 folds: the mean absolute SHAP value per genus ($|SHAP|$) was used as the importance metric for both models, and is not affected by the direction issue described below.

Direction is not defined from mean signed SHAP. For a linear model evaluated with `LinearExplainer` under interventional feature perturbation, the SHAP value for feature j is approximately $\beta_j \cdot (x_{ij} - \bar{x}_j^{\text{background}})$, so the mean signed SHAP value over an evaluation set is approximately $\beta_j \cdot (\bar{x}_j^{\text{eval}} - \bar{x}_j^{\text{background}})$ — a quantity that depends on both the fitted coefficient *and* the mean-abundance difference between the evaluation set and whichever background set was chosen, not on

the coefficient sign alone. Within a single cohort's own OOF folds this background-vs-eval difference is close to sampling noise (both come from the same cohort), but it is not zero, and for cross-cohort (LOCO) SHAP the background (the N-1 training cohorts) and evaluation set (the held-out cohort) come from systematically different populations by construction — exactly the compositional difference quantified by PERMANOVA in Section 2.6 — so mean signed LOCO SHAP is confounded with raw cross-cohort abundance differences rather than reflecting the fitted model's direction. For this reason, mean signed SHAP is not used anywhere in this paper as a global, cross-cohort direction or biological-association statistic.

Coefficient-based direction and stability (logistic regression only). For logistic regression, direction is instead defined from the fitted model's own coefficient. In each of the 10 outer OOF folds per cohort, the coefficient for every genus was recorded, together with its sign. A genus's direction in a given cohort was called **stable** — positive or negative — if the coefficient carried that sign in at least 8 of the 10 outer folds (a threshold specified before any results were examined); a coefficient of exactly zero counted toward neither direction. A genus was classified as meeting this **descriptive coefficient-sign-stability screen** only if it (a) appeared in the top-20 OOF mean-|SHAP| ranking in at least two cohorts, (b) had a stable positive coefficient in at least one of those cohorts, and (c) had a stable negative coefficient in at least one other. This is a conservative definition relative to a naive sign check on mean SHAP, because it requires the direction to be reproducible across most of the 10 outer folds within each cohort, not merely to differ in its single pooled mean-SHAP sign. **We treat this screen as descriptive, not as a formal hypothesis test: the 10 outer-CV folds within a cohort share substantially overlapping training data (each fold trains on the same 9/10 of that cohort's samples), so fold-level coefficient signs are not independent statistical replications, and no multiplicity-adjusted p-value is assigned to any individual taxon. LightGBM is a tree ensemble with no single coefficient-like, globally monotonic direction, and its SHAP feature-genus relationships may be nonlinear or non-monotonic; no coefficient-based (or any other) global direction or stability screen is therefore applied to LightGBM anywhere in this paper.** LightGBM results are limited to OOF mean-|SHAP| feature importance and the top-feature-overlap/Jaccard analysis below, which do not require a direction assignment.

For LOCO analyses, SHAP values were computed on the held-out test cohort from each LOCO-trained model, with the three-cohort training pool as the SHAP background, and are reported and used only as a feature-*importance* ranking (mean |SHAP|); consistent with the confound described above, no cross-cohort AD/CN direction is assigned to LOCO SHAP values for either model.

Cross-cohort taxonomic overlap. Pairwise taxonomic overlap between cohorts was quantified using Jaccard similarity on the top-N SHAP genera ($N \in \{10, 20, 50\}$); $N=20$ is the primary threshold. The statistic actually reported is the mean of the six unique pairwise Jaccard values among the four labeled cohorts' top-N sets ($\binom{4}{2} = 6$ pairs). The null distribution against which this statistic is compared was constructed to match it exactly: in each of 100,000 replicates, four independent random top-N gene sets were drawn (without replacement) from the full 396-genus pool, and the same six unique pairwise Jaccard similarities among those four random sets were averaged to produce one null replicate — the same statistic-construction procedure as the observed value, rather than a single pairwise Jaccard between two random sets. The empirical one-sided p-value was computed as $(1 + \# \{ \text{null} \geq \text{observed} \}) / (n_{\text{perm}} + 1)$ (Davison and Hinkley, 1997). A genus was classified as a directional flip using the coefficient-stability criterion above, not from this overlap analysis.

2.9 AI Tool Disclosure

Initial manuscript drafts and computational scripts were prepared with minimal assistance from Claude (Anthropic). All analytical decisions, parameter choices, experimental design, statistical interpretations, and scientific conclusions were made by the author. All code and text were reviewed, validated, and substantially revised under the author's direction.

2.10 Statistical Analysis Environment

DADA2 preprocessing (Section 2.2) was run using DADA2 v1.28 in R v4.3 and was not rerun for the Round 2 revision described below. All original (Round 1) Python analyses used scikit-learn v1.5, lightgbm v4, shap v0.51, pandas v2.x, and numpy v1.x in a conda environment (Python 3.11); original R analyses used vegan v2.6, sva v3.50 (ComBat-seq), and MMUPHin v1.15 in R v4.3. The Round 2 analyses added in response to peer review (training-only feature-selection sensitivity, label-blind batch correction, SHAP coefficient-stability, and the matched Jaccard null; Sections 2.5.1, 2.7, 2.8) were executed in Python 3.11.15 (scikit-learn 1.5.2, lightgbm 4.6.0, shap 0.51.0 — matching the original Python package versions) and R 4.5.1 (vegan 2.7.1, sva 3.58.0, MMUPHin 1.24.0 — newer than the original R environment, reflecting routine system updates between the two analysis rounds). This mixed provenance — original DADA2/R4.3 preprocessing feeding into a partially re-run, newer R4.5.1/unchanged-Python downstream analysis — is disclosed here rather than described as a single uniform environment. Figures were generated with matplotlib v3.8 (Round 1) or v3.11 (Round 2 SHAP/Figure 6 regeneration) at 300 DPI.

3. Results

3.1 Cohort Characteristics and Processing

Five cohorts comprising 599 deposited samples from three countries were assembled (Figure 1A; Table 1). After DADA2 processing and chimera removal, read retention rates varied substantially across cohorts (47.9%–78.3%; Supplementary Table S2), reflecting differences in sequencing platform, library preparation, and amplicon length. The Kazakhstan cohort (single-end, NovaSeq) had notably different quality dynamics compared to the paired-end MiSeq cohorts. SILVA v138.1 genus-level classification and cross-cohort union produced a unified feature matrix of 396 genera. The 90 blood microbiome (B_*) samples from Zhu 2022 were excluded prior to all analyses (Section 2.4), leaving a working dataset of 509 samples across five cohorts. CLR transformation was applied to all 509 samples jointly after union, with multiplicative zero-replacement at $\delta=0.65$.

For supervised classifier analyses (Phases 2–3 and 5–6), the Kim/KBASE 2022 cohort was excluded because per-sample diagnosis labels are IRB-restricted (Supplementary Note S1), leaving four labeled cohorts with a total of 401 samples for binary AD/CN analyses (86 Zhuang + 171 Ling + 60 Zhu fecal binary subset + 84 Kazakhstan; Zhu 2022 MCI samples excluded from binary analyses).

3.2 Within-Cohort Classification Performance

Within-cohort nested cross-validation yielded AUC-ROC values broadly consistent with those reported in the original publications (Figure 2; Table 2). Logistic regression achieved AUC-ROC of 0.633 [95% CI:

0.51–0.75] for Zhuang 2018, 0.865 [0.81–0.92] for Ling 2020, 0.998 [0.99–1.00] for Zhu 2022, and 0.766 [0.66–0.86] for Kazakhstan. LightGBM yielded similar values: 0.635 [0.52–0.76], 0.862 [0.81–0.91], 0.979 [0.95–1.00], and 0.694 [0.57–0.80] respectively.

The near-perfect within-cohort AUC for Zhu 2022 fecal samples (logistic regression 0.998; LightGBM 0.979; n=60 binary) is notable given the small sample size (30 per class). With 10-fold CV on 60 samples, each test fold contains approximately six samples; fold-level AUC estimates can be unstable at this scale. PERMANOVA analysis (Section 3.4) reveals that diagnosis explains $R^2=0.094$ of within-cohort compositional variance ($F=6.01$, $p=0.0001$), suggesting statistically detectable diagnosis-associated compositional differences in this cohort. However, the high AUC should be interpreted cautiously and the LOCO experiment (Section 3.3) provides a more informative assessment of how transferable this signal is.

The three-class analysis (AD / MCI / CN) on all 90 Zhu 2022 fecal samples using logistic regression (one-vs-rest macro-AUC under nested cross-validation) yielded macro-AUC=0.761, confirming the multi-class discriminability of this cohort’s fecal compositional data.

For Zhuang 2018, within-cohort AUC values (0.633–0.635) were modest, consistent with the small sample size and weak within-cohort diagnosis effect observed in PERMANOVA ($R^2=0.017$, $p=0.066$; Section 3.4).

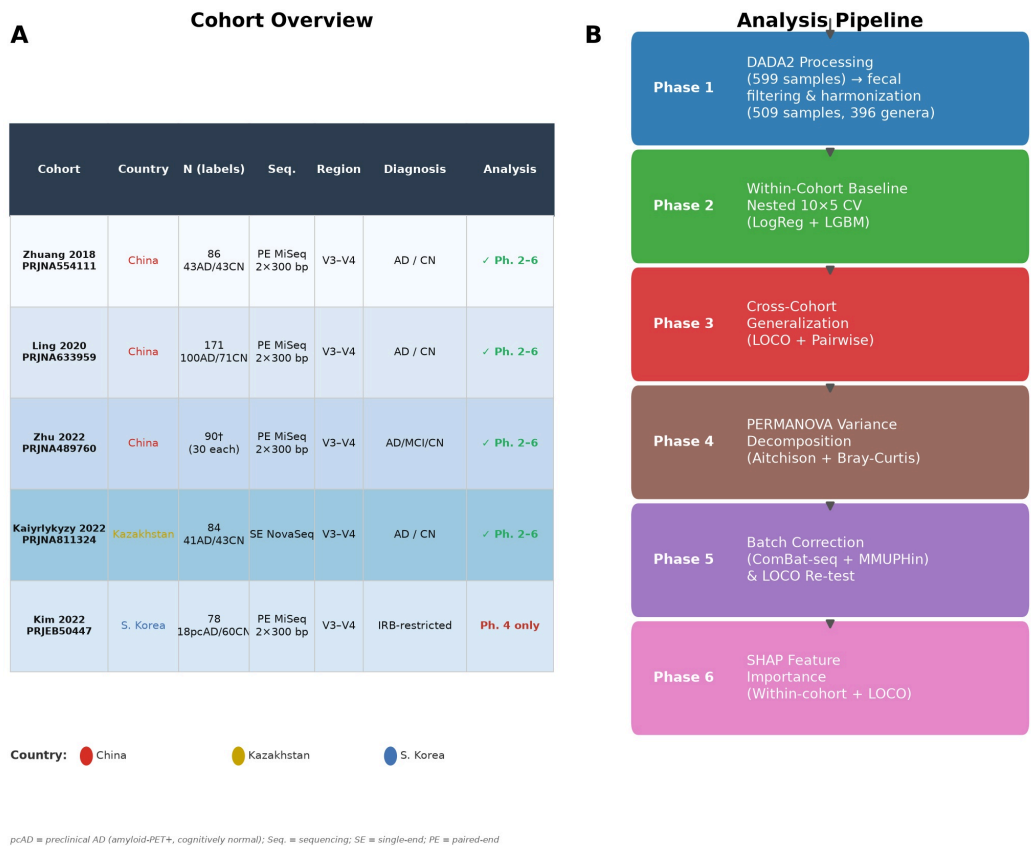
**Phase 4****Phase 5****Phase 6**

Figure 1

Figure 1. Study design. **(A)** Overview of the five 16S rRNA amplicon sequencing cohorts assembled for this study, including country of origin, sample size, sequencing mode, 16S variable region, diagnosis label availability, and analytic phases in which each cohort participates. **(B)** Summary of the six-phase analysis pipeline, from DADA2 processing through SHAP feature importance analysis.

Table 1. Cohort characteristics.

Cohort	Country	Accession	N (analyzed)	AD	CN	MCI	Sequencing	Library	Supervised Analysis
Zhuang 2018	China	PRJNA554111	86	43	43	—	MiSeq 2×300	Paired-end	Phases 2–6
Ling 2020	China	PRJNA633959	171	100	71	—	MiSeq 2×300	Paired-end	Phases 2–6
Zhu 2022†	China	PRJNA489760	90 (fecal)	30	30	30	MiSeq 2×300	Paired-end	Phases 2–6
Kaiyrlykzy 2022	Kazakhstan	PRJNA811324	84	41	43	—	NovaSeq	Single-end	Phases 2–6
Kim 2022	South Korea	PRJEB50447	78	18*	60	—	MiSeq 2×300	Paired-end	Phase 4 only
Total			509	232	247	30			

Amyloid-PET positive preclinical AD; per-sample labels IRB-restricted and not publicly available. †Zhu 2022 deposited 180 samples (90 fecal F₋ + 90 blood B₋); only fecal samples retained for analysis.*

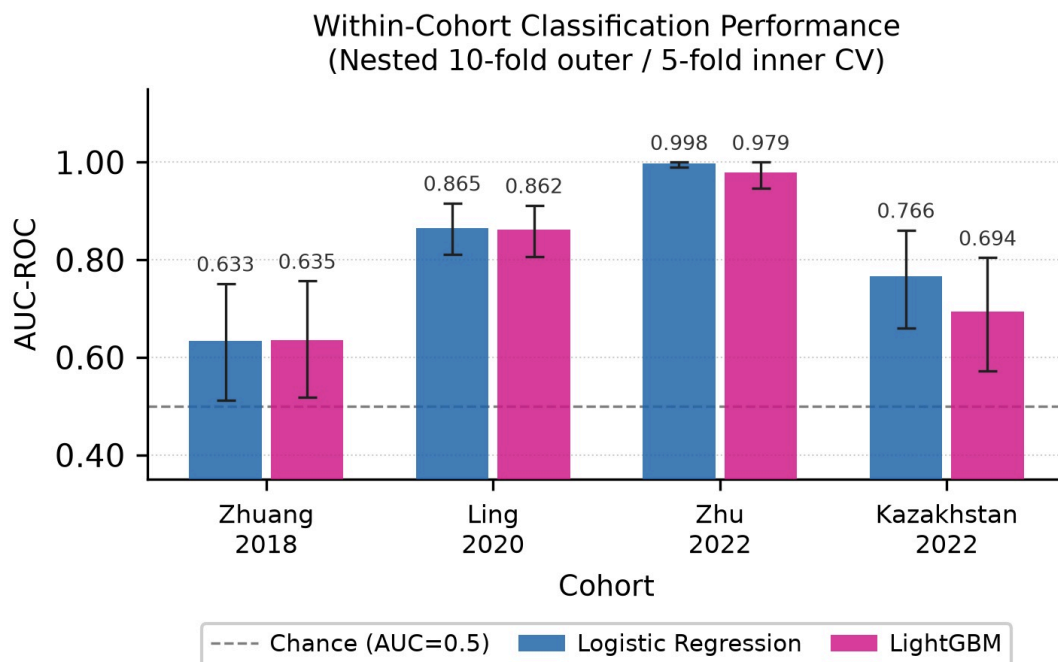
**Figure 2**

Figure 2. Within-cohort classification performance under nested cross-validation (10-fold outer / 5-fold inner CV). Bars show out-of-fold AUC-ROC; error bars represent 95% bootstrap confidence intervals (1,000 resamples). Dashed line = chance (AUC = 0.50). LogReg = L2-regularized logistic regression; LGBM = LightGBM.

Table 2. Within-cohort nested cross-validation performance (10-fold outer, 5-fold inner CV).

Cohort	Model	N	AUC-ROC	95% CI	Sensitivity	Specificity	Originally Reported AUC
Zhuang 2018	LogReg	86	0.633	[0.51–0.75]	0.814	0.512	~0.73 (Zhuang et al., 2018)
Zhuang 2018	LGBM	86	0.635	[0.52–0.76]	0.884	0.442	—
Ling 2020	LogReg	171	0.865	[0.81–0.92]	0.780	0.859	~0.87 (Ling et al., 2021)
Ling 2020	LGBM	171	0.862	[0.81–0.91]	0.740	0.887	—
Zhu 2022	LogReg	60	0.998	[0.99–1.00]	0.967	1.000	~0.94 (Zhu et al., 2022)
Zhu 2022	LGBM	60	0.979	[0.95–1.00]	0.900	1.000	—
Kaiyrlykyzy 2022	LogReg	84	0.766	[0.66–0.86]	0.829	0.605	~0.78 (Kaiyrlykyzy et al., 2022)
Kaiyrlykyzy 2022	LGBM	84	0.694	[0.57–0.80]	0.610	0.767	—

N=60 for Zhu 2022 binary analysis (30 AD + 30 CN fecal samples; MCI and blood samples excluded). Originally reported AUC values are approximate, derived from the primary publications using the best-performing classifier reported; direct numerical comparison is limited by differences in cohort subset, model type, and validation design.

3.3 Cross-Cohort Generalization Failure

LOCO cross-cohort AUC-ROC values revealed substantial and variable degradation in cross-cohort performance across all cohorts and model types (Figure 3A; Table 3). Mean LOCO AUC across the four cohorts was 0.640 for logistic regression and 0.647 for LightGBM—corresponding to mean drops of 0.176 and 0.146 AUC points respectively from within-cohort performance. One of eight model-cohort combinations produced LOCO AUC near chance: Zhuang 2018 with logistic regression (AUC=0.504 [0.37–0.63]).

The largest absolute logistic regression AUC drop was observed for Kazakhstan (within-cohort 0.766 → LOCO 0.564, $\Delta=0.202$). Zhu 2022 showed the largest LightGBM drop: within-cohort 0.979 → LOCO 0.758 ($\Delta=0.221$); its logistic regression drop was also substantial (0.998 → 0.813, $\Delta=0.185$). These results are consistent with LOCO models relying on predictive patterns learned from training cohorts that do not transfer fully to the held-out cohort, where cohort-specific compositional structure may contribute to within-cohort performance.

Ling 2020 also experienced substantial LOCO collapse: logistic regression 0.865 → 0.678 ($\Delta=0.187$); LightGBM 0.862 → 0.674 ($\Delta=0.188$). LightGBM confidence intervals for the Ling 2020 LOCO estimate [0.59–0.75] do not overlap with the within-cohort estimate [0.81–0.91], suggesting a performance drop unlikely to be explained by sampling variability alone.

LOCO performance was heterogeneous across cohorts: Zhu 2022 achieved the highest LOCO AUC (logistic regression 0.813), while Zhuang 2018 collapsed to near chance. It should be noted that only one

of the eight main model–cohort combinations fell to near chance, and five of eight exceeded 0.55; the results therefore demonstrate substantial and variable degradation rather than uniform failure.

Sensitivity analyses confirmed that generalization failure was not attributable to Kazakhstan, Zhuang 2018, or Ling 2020 individually (Supplementary Figure S4). When Kazakhstan was excluded and LOCO was performed on the three remaining Chinese cohorts, Zhuang 2018 remained near chance (LogReg 0.512, LightGBM 0.567), demonstrating that the generalization failure for Zhuang 2018 is not a Kazakhstan artifact. When Zhuang 2018 was excluded, LOCO on the remaining three cohorts still showed AUC values that were substantially below within-cohort baselines for Kazakhstan (LogReg 0.599, LightGBM 0.495). When Ling 2020 was excluded, Kazakhstan LOCO LogReg was 0.604. These sensitivity results confirm the robustness of the main finding across cohort compositions.

Training-only feature-selection sensitivity (Section 2.5.1) confirmed that this LOCO generalization-failure pattern is not an artifact of the global, all-cohort genus-selection step (Section 2.2). Re-deriving the retained genus set independently within each split — using only that split’s training-side samples, with the held-out cohort contributing zero information to feature selection — changed LOCO AUC by at most 0.066 in either direction (largest changes: Kazakhstan LightGBM 0.591→0.525, $\Delta=-0.066$; Kazakhstan LogReg 0.564→0.609, $\Delta=+0.045$; all other cohort–model combinations changed by ≤ 0.031). The retained genus count under strict training-only selection ranged from 150 (when Kazakhstan — a comparatively high-prevalence cohort for many genera — was excluded from feature selection during its own hold-out) to 373, versus 396 under the global rule, yet no cohort–model combination changed its qualitative classification (near-chance, intermediate, or robust). Within-cohort nested-CV AUC under training-only selection likewise moved by -0.083 to $+0.038$ (Supplementary Table S4). We conclude that the global feature-selection design in the main analysis, while a genuine methodological limitation in principle, did not materially inflate or deflate the magnitude of the reported cross-cohort generalization failure.

Table 3. LOCO cross-cohort AUC-ROC versus within-cohort nested CV baseline.

Test Cohort	Model	Within-Cohort AUC	LOCO AUC	LOCO 95% CI	AUC Drop
Zhuang 2018	LogReg	0.633	0.504	[0.37–0.63]	0.129
Zhuang 2018	LGBM	0.635	0.565	[0.44–0.69]	0.070
Ling 2020	LogReg	0.865	0.678	[0.59–0.76]	0.187
Ling 2020	LGBM	0.862	0.674	[0.59–0.75]	0.188
Zhu 2022	LogReg	0.998	0.813	[0.69–0.92]	0.185
Zhu 2022	LGBM	0.979	0.758	[0.63–0.87]	0.221
Kazakhstan	LogReg	0.766	0.564	[0.44–0.69]	0.202
Kazakhstan	LGBM	0.694	0.591	[0.47–0.71]	0.103
Mean	LogReg	0.816	0.640	—	0.176
Mean	LGBM	0.793	0.647	—	0.146

Pairwise single-cohort transfer AUC values (Figure 3B–C; Supplementary Table S1) ranged from 0.54 to 0.88 for logistic regression. The highest pairwise AUC was Ling 2020 → Zhu 2022 (logistic regression AUC=0.88), reflecting transfer between two large Chinese fecal cohorts with overlapping compositional structure. Kazakhstan → Ling 2020 also transferred reasonably (logistic regression AUC=0.74). Most

other pairwise AUCs clustered between 0.54 and 0.70, consistent with the overall pattern of limited cross-cohort transferability.

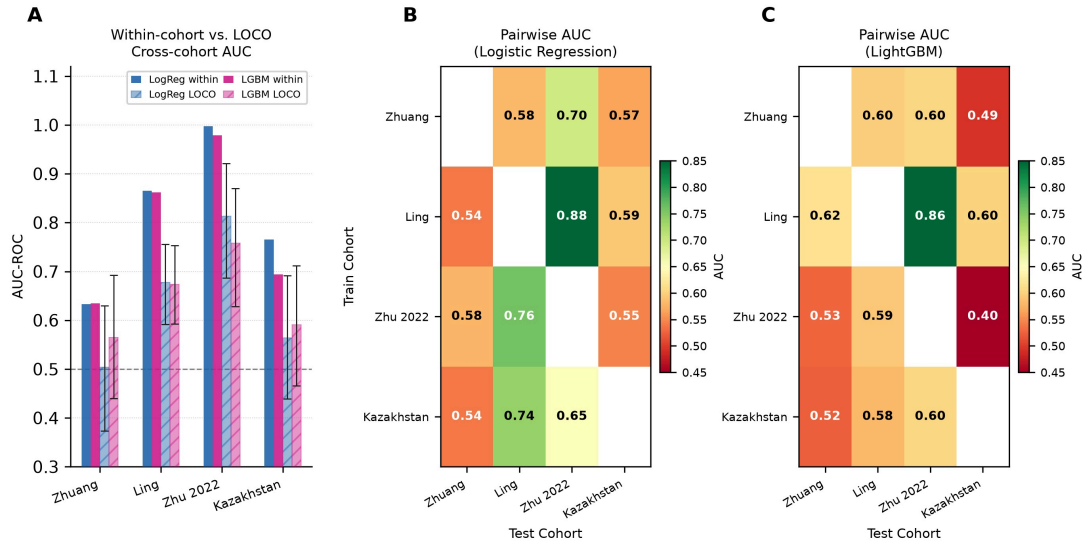


Figure 3

Figure 3. Cross-cohort generalization failure. (A) Within-cohort (solid bars) versus LOCO cross-cohort (hatched bars) AUC-ROC for logistic regression (blue) and LightGBM (pink). Error bars = 95% bootstrap CI. (B, C) Pairwise single-cohort-train / single-cohort-test AUC heatmaps for logistic regression and LightGBM respectively. Diagonal (within-cohort) values are omitted; colour scale: red = near chance, green = high AUC.

3.4 PERMANOVA Variance Decomposition

PERMANOVA on Aitchison distance across all five cohorts showed that cohort identity explained 19.3% of total compositional variance ($R^2=0.193$, $F=30.13$, $p=0.0001$, 9,999 permutations; Figure 4A). On Bray-Curtis dissimilarity, the cohort R^2 was 0.136 ($F=19.80$, $p=0.0001$). The cohort effect was substantially larger than disease signal: in a marginal two-variable model restricted to the four labeled cohorts ($n=401$), cohort (marginal) explained $R^2=0.172$ ($F=28.03$, $p=0.0001$) versus diagnosis (marginal) $R^2=0.014$ ($F=6.82$, $p=0.0001$)—an approximately 12-fold disparity within the same model. This is an empirical statement about the relative magnitude of cohort- versus diagnosis-associated compositional heterogeneity in these data. It provides context for interpreting the batch-correction results in Section 3.5, but PERMANOVA marginal R^2 values do not by themselves establish that the cohort and diagnosis axes of variation overlap, nor do they explain why any particular batch-correction algorithm changed classifier performance in a specific direction; that relationship is discussed as a separate, more limited hypothesis in Section 4.2.

A logistic regression classifier trained to predict cohort identity from the CLR-transformed features across all 509 samples achieved macro-AUC=0.9917 (one-vs-rest, stratified 5-fold CV; per-cohort AUC: Kazakhstan 1.000, Kim 2022 0.997, Ling 2020 0.993, Zhuang 2018 0.989, Zhu 2022 0.980). Cohort membership is thus near-perfectly recoverable from the same features used for disease classification, confirming that cohort-specific information dominates the compositional feature space.

Principal coordinates analysis (PCoA) on Aitchison distance (Figure 4B; $n=509$) visually confirmed strong cohort-level clustering, with partial overlap between Chinese cohorts (Zhuang 2018, Ling 2020) but clear separation of Kazakhstan and Zhu 2022. The Kim/KBASE 2022 cohort was clearly separated from all

others along PC1, consistent with its preclinical amyloid-based staging, South Korean origin, and potentially distinct dietary patterns.

Beta-dispersion testing revealed significant heterogeneity in within-group spread across cohorts ($F=13.93$, $p=0.0001$, 9,999 permutations; Figure 4C), meaning that the PERMANOVA R^2 of 0.193 reflects both compositional centroid shifts and differences in within-cohort spread. Kazakhstan showed the greatest within-cohort variance, while the Chinese cohorts (Zhuang, Ling) were more tightly dispersed. This dispersion heterogeneity is relevant to interpreting PERMANOVA effect sizes and is acknowledged in the Methods section.

Within-cohort PERMANOVA revealed highly variable disease-associated compositional effects: Ling 2020 ($R^2=0.053$, $F=9.53$, $p=0.0001$) and Zhu 2022 ($R^2=0.094$, $F=6.01$, $p=0.0001$) showed significant diagnosis effects, while Zhuang 2018 ($R^2=0.017$, $p=0.066$) and Kazakhstan ($R^2=0.022$, $p=0.069$) did not reach statistical significance at $\alpha=0.05$. These within-cohort disease effect sizes broadly rank-order with within-cohort AUC performance, providing a triangulating validation of the classifier results. The Zhu 2022 fecal-only $R^2=0.094$ represents the highest within-cohort disease effect among the four labeled cohorts, consistent with a detectable diagnosis-associated compositional difference in this dataset.

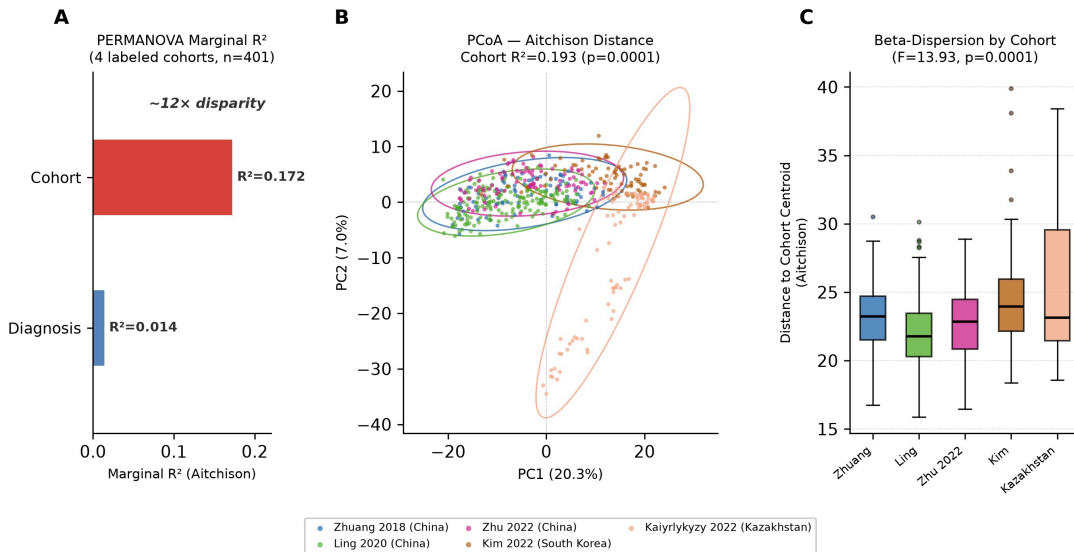


Figure 4

Figure 4. PERMANOVA variance decomposition and beta-diversity structure. **(A)** Marginal R^2 values from a two-variable PERMANOVA model on the four labeled cohorts ($n=401$): cohort $R^2=0.172$ and diagnosis $R^2=0.014$, shown as independent bars. These Type III marginal R^2 values are estimated independently and do not sum to 1. **(B)** PCoA of Aitchison distance matrix (all five cohorts; $n=509$). Cohort $R^2=0.193$ ($F=30.13$, $p=0.0001$, 9,999 permutations). Ellipses = 95% normal confidence ellipses. **(C)** Beta-dispersion: distance from each sample to its cohort centroid ($F=13.93$, $p=0.0001$, 9,999 permutations).

3.5 Batch Correction Does Not Recover Cross-Cohort Generalization

Batch correction was applied using ComBat-seq (on raw counts) and MMUPHin (on relative abundances) to the four labeled cohorts with cohort as the batch variable, under the two designs described in Section 2.7: a **label-blind** design (no diagnosis information used during correction; primary analysis) and a **label-informed** design (each sample's own true diagnosis label used during correction; exploratory/oracle-style sensitivity only) (Figure 5; Table 4).

Table 4. Mean LOCO AUC across four cohorts, by correction design and model.

Design	Method	LogReg Mean LOCO AUC	LogReg Mean Drop from Within-Cohort	LGBM Mean LOCO AUC	LGBM Mean Drop
—	Uncorrected	0.640	0.176	0.647	0.146
Label-blind (primary)	ComBat-seq	0.634	0.182	0.654	0.139
Label-blind (primary)	MMUPHin	0.639	0.176	0.648	0.145
Label-informed (exploratory)	ComBat-seq	0.493	0.322	0.555	0.238
Label-informed (exploratory)	MMUPHin	0.612	0.203	0.582	0.210

Label-blind correction (primary result): batch correction leaves cross-cohort generalization essentially unchanged. Under label-blind ComBat-seq and MMUPHin — the design that does not use any sample’s diagnosis label — mean LOCO AUC changed by no more than 0.007 in either direction relative to uncorrected, for either model (Table 4). No individual cohort collapsed: per-cohort logistic-regression LOCO AUC under label-blind ComBat-seq ranged from 0.525 (zhuang2018) to 0.769 (shanghai2022/Zhu 2022), closely tracking the uncorrected values (0.504–0.813) rather than diverging sharply from them. **Batch correction, evaluated under a design that cannot see the answer it is being asked to help predict, neither improves nor substantially worsens cross-cohort transferability in this dataset — it is simply ineffective at recovering it.**

Label-informed correction (exploratory/oracle-style sensitivity; Supplementary Figure S5): the previously reported severe degradation is largely an artifact of using target labels during correction, not evidence about batch correction’s effect on biological signal. Under the label-informed design — which uses every sample’s own true diagnosis label during correction and therefore cannot represent a deployable or even a fair oracle-style comparison — ComBat-seq reduced mean logistic-regression LOCO AUC from 0.640 to 0.493, with the Kazakhstan cohort collapsing to AUC=0.308 [0.20–0.42]. The same cohort under label-blind ComBat-seq scored 0.560, closely matching its uncorrected value (0.564). This side-by-side comparison indicates that the severe, cohort-specific collapse previously observed was driven substantially by the label-informed design itself — most likely because a global correction model that is explicitly told each sample’s own diagnosis label (via `full_mod=TRUE/covariates="diagnosis"`) constructs a label-conditional transformation that does not correspond to any transformation a genuinely prospective classifier could apply, rather than by an intrinsic incompatibility between ComBat-seq’s negative-binomial framework and 16S compositional data. MMUPHin’s label-informed effect was milder (LogReg mean: 0.640→0.612) but still did not improve on label-blind or uncorrected performance.

Within-cohort AUC on label-blind-corrected data was similar to uncorrected (within ± 0.05 for most cohort–method combinations; Supplementary Table S5), which is a fair, unconfounded comparison because the label-blind correction never saw any sample’s diagnosis label. **We do not draw the same conclusion from within-cohort AUC on label-informed-corrected data — that comparison is confounded,** because the label-informed correction’s own covariate-preservation mechanism (`full_mod=TRUE` for ComBat-seq; `covariates="diagnosis"` for MMUPHin) is explicitly

designed to retain diagnosis-associated structure using every sample’s own true label; any preserved within-cohort separability under that design cannot be attributed to correction-independent biological signal versus the correction algorithm’s own label-preservation step. We therefore no longer draw the previously stated conclusion that “within-cohort AUC preservation rules out indiscriminate signal removal” from the label-informed results; that inference required an unconfounded comparison that only the label-blind design provides, and the label-blind within-cohort comparison does support it descriptively (Supplementary Table S5), without establishing this as strong biological evidence beyond the observed AUC values themselves.

The empirical PERMANOVA finding that cohort explains approximately 12 times more compositional variance than disease status (marginal R^2 : 0.172 vs. 0.014; Section 3.4) provides context for why a correction method targeting cohort-associated variance might struggle to leave disease-associated variance fully intact, but it does not, by itself, establish that the two sources of variation overlap, nor does it explain the specific pattern observed here — namely that a properly label-blind correction left performance essentially flat while a label-informed one produced large, method- and cohort-specific movements. That specific pattern is more directly explained by the transductive, label-informed design itself (Section 2.7) than by the general cohort/diagnosis variance disparity, and we present it as such rather than as a demonstration that linear batch correction “cannot” separate cohort from disease signal in general.

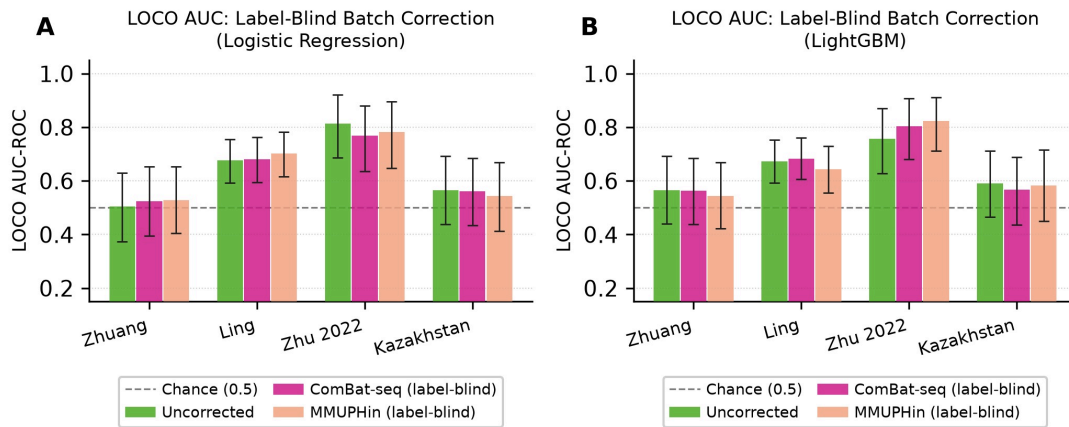


Figure 5

Figure 5. Effect of label-blind batch correction on LOCO cross-cohort AUC-ROC (primary analysis; no diagnosis information used during correction). (A) Logistic regression and (B) LightGBM LOCO AUC for each held-out cohort under three conditions: uncorrected (green), ComBat-seq label-blind corrected (magenta), and MMUPHin label-blind corrected (salmon). Error bars = 95% bootstrap CI. Dashed line = chance (0.50). Under this label-blind design, both methods left mean LOCO AUC essentially unchanged relative to uncorrected (within 0.007 AUC for both methods and models), neither recovering nor substantially worsening cross-cohort generalization. An exploratory label-informed (oracle-style) comparison, in which correction used each sample’s own true diagnosis label and which produced substantially different, more severe degradation for some cohorts, is reported separately in Supplementary Figure S5 and is not treated as evidence about prospective performance (Section 2.7, Section 3.5).

3.6 SHAP Analysis Reveals Cohort-Specific Taxonomic Signatures and a Descriptive Coefficient-Stability Screen

To characterize the feature-level patterns underlying generalization failure, we computed out-of-fold SHAP values for within-cohort classifiers and LOCO-trained classifiers (Figure 6). Using an OOF design avoids the in-sample overfitting bias identified in prior SHAP analyses of microbiome classifiers, where

training-set SHAP values can reflect model memorization rather than the model's out-of-sample feature importance. Feature importance (mean |SHAP|) is reported for both models; direction is reported only for logistic regression, using the fitted-coefficient-based definition in Section 2.8 (not mean signed SHAP).

Minimal overlap in top predictive taxa across cohorts, now assessed against a null matched to the observed statistic. Pairwise Jaccard similarity of the top-20 SHAP genera between cohort pairs averaged 0.135 for logistic regression and 0.058 for LightGBM (Figure 6C) — the mean of the six unique pairwise Jaccard values among the four labeled cohorts. Against a null distribution constructed to match this statistic exactly (100,000 replicates, each averaging the six pairwise Jaccards among four random top-20 sets; Section 2.8), logistic regression overlap was significantly above chance (null mean 0.027 [95% CI: 0.009–0.049]; $p < 0.0001$), and LightGBM overlap was also significantly above chance at this N ($p = 0.0048$). A sensitivity check across $N \in \{10, 20, 50\}$ (Supplementary Table S7) shows logistic regression significant at all three thresholds, while LightGBM was significant at $N = 20$ and $N = 50$ but not at $N = 10$ ($p = 0.45$) — indicating that LightGBM's top-predictor overlap across cohorts, while small in absolute magnitude (0.058, versus a null expectation of 0.027), is a real, detectable signal at the primary and larger N thresholds, more consistent than previously reported but still substantially smaller than logistic regression's overlap and correspondingly less informative for identifying a small set of consistently important taxa.

Cohort-specific top predictors, and a concrete illustration of why coefficient-based direction was needed. The most discriminative genus for logistic regression was *Subdoligranulum* in Zhuang 2018, while *Akkermansia* was the top predictor in the remaining three cohorts (Zhu 2022 |SHAP|=0.228; Kazakhstan |SHAP|=0.136; Ling 2020 |SHAP|=0.270). Under the fitted model's coefficient, *Akkermansia* has a **stable positive coefficient in all four cohorts** (10 of 10 outer folds positive in every cohort; Supplementary Table S6) — higher CLR abundance consistently predicts higher AD log-odds under the fitted model, with no coefficient-based flip anywhere. This is a useful negative control for the direction-definition problem itself: the pooled mean SHAP value for this same genus is negative in Ling 2020 and Zhuang 2018 (which an earlier, mean-SHAP-based version of this analysis read as “CN-associated”) but positive in Zhu 2022 and Kazakhstan — an apparent flip that does **not** reflect the coefficient the model actually learned, and is instead a background-relative artifact of the kind described in Section 2.8. We report this discrepancy explicitly because it demonstrates, on real data from this study, why mean signed SHAP was not an adequate direction statistic and why the coefficient-based definition is used for every direction claim in this paper. For LightGBM, the top taxon per cohort was entirely distinct: *Lachnoclostridium* (Zhuang 2018), *Akkermansia* (Ling 2020), *Bacteroides* (Zhu 2022), *Castellaniella* (Kazakhstan) (Figure 6A; Supplementary Figure S2); no direction is assigned to these LightGBM values (Section 2.8).

Coefficient-sign stability screen (logistic regression only). Using the pre-specified coefficient-stability criterion (Section 2.8: a stable positive coefficient sign in ≥ 8 of 10 outer folds in at least one cohort, and a stable negative coefficient sign in ≥ 8 of 10 outer folds in at least one other cohort, restricted to genera in the top-20 OOF mean-|SHAP| ranking in ≥ 2 cohorts), **eight genera met this descriptive stability criterion** (Figure 6B): *Agathobacter*, *Bifidobacterium*, *Coprococcus*, *Dorea*, *Lactobacillus*, *NK4A214 group*, *Romboutsia*, and *Ruminococcus gnavus group*. We describe this deliberately as a descriptive screen rather than a statistically confirmed set of contradictions: the 10 outer-CV folds share substantially overlapping training sets (each fold's training data is 9/10 of the same cohort), so fold-level coefficient signs are not 10 independent replications, and no multiplicity-adjusted p-value is assigned to any

individual taxon (Section 2.8, Limitations). This is nonetheless a smaller and more conservative set than an earlier, less rigorous analysis of this dataset that classified any genus with opposite-signed pooled mean SHAP across two cohorts as a “flip” (which had identified twelve such genera); the stability requirement removes genera whose apparent flip was not reproducible across most of a cohort’s own outer folds. **No comparable screen is applied to LightGBM:** a tree ensemble has no single coefficient-like, globally monotonic direction, and LightGBM’s SHAP–feature relationships may be nonlinear or non-monotonic, so no scalar direction statistic — descriptive or otherwise — is computed or reported for it.

Romboutsia shows the cleanest population-level pattern among the eight taxa meeting the screen: a stable negative coefficient (higher relative abundance predicts lower AD log-odds under the fitted model) in all three Chinese cohorts (Zhuang 2018, Ling 2020, Zhu 2022) and a stable positive coefficient in Kazakhstan (9/10 folds; Supplementary Table S6) — the one genus among the eight whose pattern cleanly separates by country in this dataset. We do not have a specific mechanistic or dietary hypothesis for *Romboutsia* and do not offer one here.

Lactobacillus also meets the screen, and is a genus with a well-documented, biologically plausible reason to differ by population — it is the dominant organism in fermented dairy products (kumis, kefir) that are culturally prominent in Central Asian diets — but its actual data pattern does not cleanly support a simple Chinese-cohorts-versus-Kazakhstan narrative: it is stable-negative in Zhuang 2018 only, and stable-positive in Ling 2020, Zhu 2022, and Kazakhstan (Supplementary Table S6) — three of the four cohorts share the same direction, with Zhuang 2018 the outlier rather than Kazakhstan. We flag this discrepancy explicitly rather than fitting the fermented-dairy narrative to a pattern the data do not actually show; a plausible dietary mechanism does not by itself establish that it explains this specific empirical result.

Akkermansia is a useful contrast to both: despite appearing to reverse direction under the old, invalid mean-signed-SHAP statistic (Section 3.6), its fitted coefficient is stable and **positive in all four cohorts** (10/10 outer folds in every cohort; Supplementary Table S6) — it does not meet the screen at all under the coefficient-based definition. This genus’s abundance is strongly influenced by host diet, particularly fiber and polyphenol intake (Plovier et al., 2017), and by geographic-population-specific dietary patterns that likely modulate its abundance independently of any AD-specific effect; the coefficient-based analysis indicates that, at least under this fitted model, its association with AD status is directionally consistent across these four populations even though its raw pooled-SHAP appearance was not. All dietary explanations offered in this section remain speculative in the absence of dietary co-variate data and should be read as hypotheses, not established conclusions.

LOCO SHAP analysis: feature importance only, no cross-cohort direction assigned. When SHAP values were computed for LOCO-trained classifiers applied to held-out cohorts, *Akkermansia* was the top-importance predictor for two of four held-out cohorts — Zhuang 2018 ($|\text{SHAP}|=0.470$) and Zhu 2022 ($|\text{SHAP}|=0.372$) — while *Subdoligranulum* ($|\text{SHAP}|=0.269$) topped held-out Ling 2020 and *Christensenellaceae R-7 group* ($|\text{SHAP}|=0.368$) topped held-out Kazakhstan. No genus ranked first across all four held-out conditions. Consistent with Section 2.8, we do not assign an AD/CN direction to these LOCO SHAP values: the LOCO SHAP background (the three training cohorts) and evaluation set (the held-out cohort) are drawn from systematically different populations by construction, so the sign of a LOCO mean SHAP value is confounded with the raw cross-cohort compositional difference quantified by PERMANOVA (Section 3.4) rather than reflecting a stable, model-derived direction. The absence of a single top-importance genus across all four held-out cohorts is itself consistent with the classifier

activating whichever genera most closely resemble its training-cohort signature rather than a disease marker that generalizes across populations.

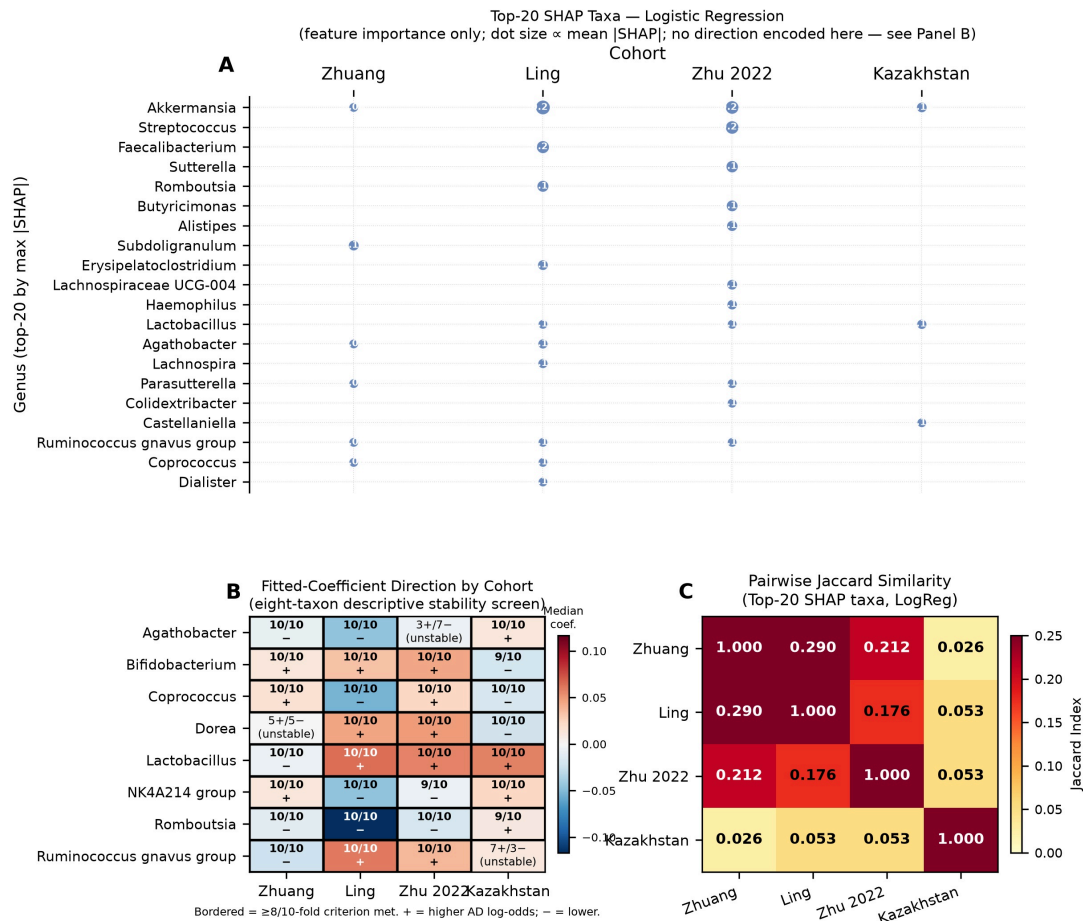


Figure 6

Figure 6. SHAP feature importance and coefficient-based direction analysis, logistic regression. **(A)** Top-20 SHAP genera by maximum |SHAP| across cohorts. Dot size $\propto$ mean |SHAP|; feature importance only — no direction is encoded in this panel. **(B)** Fitted-coefficient direction by cohort for the eight genera meeting a pre-specified descriptive coefficient-sign-stability screen (top-20 in ≥ 2 cohorts; a stable positive coefficient sign in $\geq 8/10$ outer folds in ≥ 1 cohort and a stable negative sign in $\geq 8/10$ folds in ≥ 1 other). Cell color/value = median fitted logistic-regression coefficient across the 10 outer folds (positive $\rightarrow$ higher predicted AD log-odds; negative $\rightarrow$ lower); cell text = the fold count backing that direction; bordered cells meet the $\geq 8/10$ stability criterion. This is a descriptive screen, not a multiplicity-corrected hypothesis test. **(C)** Pairwise Jaccard similarity of top-20 SHAP taxa between cohort pairs; mean of the six unique pairwise values = 0.135, significantly above a null matched to this statistic's construction (100,000 replicates of four random top-20 sets; $p < 0.0001$).

4. Discussion

4.1 Systematic Quantification of Generalization Failure

This study provides a systematic, quantitative analysis of cross-cohort generalization failure in gut microbiome-based AD classifiers. Across four labeled cohorts, two model types, two cross-cohort evaluation designs, and two batch correction methods, we consistently observe the same result: classifiers trained within one population fail to generalize reliably to another. The mean LOCO AUC drop of 0.176 points for logistic regression and 0.146 for LightGBM, with one of eight model-cohort combinations near

chance, demonstrates that previously reported within-cohort performance figures cannot be taken as evidence of a generalizable diagnostic signal.

This finding is consistent with the broader literature. The *Bacteroides* directionality reversal documented by Jemimah et al. (2023) across North American and Chinese cohorts, the contradictory reports for *Akkermansia* (Liu et al., 2020), and the highly cohort-specific patterns reported in individual papers all suggest population-level heterogeneity. A recent multi-cohort integration study (Wang et al., 2026, mSystems) similarly identified cross-geographic heterogeneity as a major obstacle; the present analysis complements that work by providing a systematic quantification of how severely within-cohort classifiers fail when applied to external populations, and by demonstrating that on average, neither method recovered cross-cohort transferability.

4.2 Batch Correction Does Not Recover Generalization, and the Reason Requires Separating an Empirical Result from a Hypothesis

Two distinct findings must be kept separate here, because conflating them was a limitation of how this section was framed in an earlier version of this manuscript.

Empirical result: cohort identity explains substantially more compositional variance than disease status (Aitchison $R^2=0.193$ across all five cohorts; marginal $R^2=0.172$ cohort vs. 0.014 diagnosis in the four-cohort two-variable model — an approximately 12-fold disparity), and cohort membership is near-perfectly recoverable from the same CLR features (macro-AUC=0.9917). This establishes that compositional space is overwhelmingly organized by cohort-of-origin. Separately and equally empirically: batch correction, evaluated under a label-blind design that cannot see any sample’s diagnosis label, left mean LOCO AUC essentially unchanged (within 0.007 AUC of uncorrected for both ComBat-seq and MMUPHin, both models; Section 3.5) — it neither recovered cross-cohort transferability nor substantially degraded it.

Hypothesis, clearly flagged as such: the much larger cohort-associated variance component provides plausible context for *why* correction targeting cohort-associated variance might struggle to isolate and remove only that variance without also touching disease-associated variance, if the two are not orthogonal in this compositional space. PERMANOVA marginal R^2 values do not establish that overlap directly, and we did not run an experiment (e.g., a simulation with a known, controlled degree of cohort/diagnosis-axis overlap) capable of testing it in this dataset. We therefore present it as a hypothesis consistent with the observed pattern, not as a demonstrated mechanism, and we no longer state that batch correction “cannot” work in general or that this analysis proves cohort and disease axes overlap.

A previous version of this analysis additionally reported that ComBat-seq’s correction “substantially worsened” LOCO AUC (mean LogReg: 0.640→0.493, with one cohort collapsing to AUC=0.308) and interpreted this as evidence that ComBat-seq’s negative-binomial framework is poorly calibrated for compositional 16S data. That result was obtained under the label-informed design, in which ComBat-seq and MMUPHin were both told every sample’s own true diagnosis label during a single joint fit across all four labeled cohorts (Section 2.7). Repeating the identical comparison under a label-blind design — same cohorts, same raw features, same downstream LOCO classifiers, the only change being that no diagnosis label is supplied to the correction step — reproduces essentially flat LOCO AUC instead of severe, cohort-specific collapse (Section 3.5, Table 4). This indicates that the earlier “ComBat-seq substantially worsens generalization” claim was driven in large part by the label-informed design itself, not by an intrinsic

incompatibility between ComBat-seq’s modeling framework and 16S compositional data; we retract that specific mechanistic claim and retain the label-informed numbers only as an explicitly-labeled exploratory sensitivity illustrating what a label-aware, non-deployable correction can do (Section 3.5).

The finding that survives this correction — and that we consider the more robust and more useful result for the field — is narrower than originally stated: under a fair, label-blind evaluation, two widely used batch-correction methods designed for other data types (RNA-seq counts; general microbiome meta-analysis) did not recover cross-cohort transferability for AD classification in this dataset. This is a demonstration of ineffectiveness at the specific task of restoring cross-cohort generalization, not a demonstration that these methods actively destroy signal, nor an explanation of the underlying compositional-geometry reason such correction is difficult. The field may still need fundamentally different approaches — multi-cohort training with cohort as an explicit stratification variable, domain adaptation methods, or meta-analytic frameworks that explicitly model between-cohort heterogeneity rather than attempting to remove it — but we no longer claim to have shown mechanistically why standard correction fails, only that it does not succeed under the conditions tested here.

4.3 Implications of the Zhu 2022 Paired-Sample Design

The Zhu 2022 cohort deposited both blood microbiome (B_) and fecal (F_) samples from the same 90 participants (PRJNA489760). The blood samples were excluded from this analysis because blood and fecal microbiome communities represent fundamentally distinct compartments; including both would introduce a sample-type confound identical in structure to a batch effect, inflating within-cohort AUC through tissue-compartment discrimination rather than disease discrimination.

With fecal-only samples (n=60 binary), the near-perfect within-cohort AUC (logistic regression 0.998, LightGBM 0.979) is more difficult to attribute to a single confound. The small sample size (30 per class) is relevant: with 10-fold CV on 60 samples, each test fold contains approximately 6 samples, and single-fold AUC estimates can be highly variable. Zhu 2022 showed the largest detectable diagnosis-associated compositional effect ($R^2=0.094$, $F=6.01$, $p=0.0001$) of the four labeled cohorts, but an AUC of 0.998 likely still reflects optimistic estimation at this sample size.

The LOCO result for Zhu 2022 (logistic regression AUC=0.813, the highest LOCO value among all cohorts) is informative: classifiers trained on the other three cohorts transfer to Zhu 2022 fecal samples better than to any other cohort. This suggests that the fecal disease signal in this cohort is somewhat consistent with signals in the Chinese and Kazakh training cohorts, unlike the spurious compartment-discrimination signal that would have characterized the original mixed dataset. We recommend that future AD microbiome studies report sample types explicitly and analyze each compartment independently before pooling.

4.4 Coefficient-Sign Stability Screen: A Conservative, Descriptive View of Cross-Population Contradictions

Using a pre-specified coefficient-sign-stability criterion (≥ 8 of 10 outer folds agreeing in sign; Section 2.8), eight genera met this descriptive screen for logistic regression: *Agathobacter*, *Bifidobacterium*, *Coprococcus*, *Dorea*, *Lactobacillus*, *NK4A214 group*, *Romboutsia*, and *Ruminococcus gnavus group*. This is a smaller, more conservative set than the meta-analytic literature’s qualitative suggestion that some taxa carry opposite associations with AD status in different populations might imply, and we present it

descriptively rather than as a statistically confirmed set of contradictions: the 10 outer-CV folds share substantially overlapping training data (each fold’s training set is 9/10 of the same cohort), so fold-level coefficient signs are not independent replications in the formal statistical sense, and we do not assign multiplicity-adjusted p-values to individual taxa (Section 2.8, Limitations). We emphasize “conservative” deliberately — an earlier version of this analysis defined a flip as any genus with opposite-signed *pooled mean SHAP* across two cohorts (a background-relative quantity, not the model’s coefficient; Section 2.8) and reported twelve such genera for logistic regression and, using the same invalid statistic, a further twelve for LightGBM. Requiring the direction to be reproducible across most of a cohort’s own outer folds, and grounding it in the fitted coefficient rather than a background-dependent SHAP quantity, reduces the logistic-regression count to eight; no comparable count is reported for LightGBM at all, because a tree ensemble has no single coefficient-like global direction to test for stability in the first place (Section 2.8). We treat the removal of the LightGBM claim as a correction, not merely a footnote: the original twelve-taxon LightGBM “flip” list should not be relied upon or cited as evidence of directional inconsistency in tree-based models from this dataset.

Romboutsia shows the cleanest population-level pattern among the eight: a stable negative fitted coefficient (higher relative abundance predicts lower AD log-odds) in all three Chinese cohorts, and a stable positive coefficient in Kazakhstan. We do not have a specific mechanistic or dietary hypothesis for this genus. *Lactobacillus* also meets the screen and has an available biological rationale — it is the dominant organism in fermented dairy products (kumis, kefir) culturally prominent in Central Asian diets — but its actual per-cohort pattern is messier than a clean population split: stable-negative in Zhuang 2018 only, and stable-positive in Ling 2020, Zhu 2022, and Kazakhstan (Supplementary Table S6). We report this pattern as observed rather than fitting the fermented-dairy narrative to a Chinese-versus-Kazakhstan split the data do not show; a biologically plausible mechanism for a genus does not by itself establish that the mechanism explains this dataset’s specific result.

Akkermansia illustrates why the coefficient-based, stability-gated definition matters, in the clearest possible way: its pooled mean SHAP value appears to reverse sign between Ling 2020/Zhuang 2018 and Zhu 2022/Kazakhstan (Section 3.6), which an earlier, mean-SHAP-based version of this analysis read as a directional flip — but its fitted logistic-regression coefficient is positive and stable (10 of 10 outer folds) in all four cohorts (Supplementary Table S6), meaning the model never actually learned an inconsistent direction for this genus; the apparent flip was an artifact of the background-relative SHAP statistic, not a property of the fitted model. *A. muciniphila* abundance is strongly influenced by host diet, particularly fiber and polyphenol intake (Plovier et al., 2017), and geographic-population-specific dietary patterns likely modulate its abundance independently of any AD-specific effect; under the coefficient-based analysis, this genus’s association with AD status is directionally consistent (not flipped) across all four populations studied here.

The critical implication of the eight taxa meeting this descriptive screen is that they may reflect biological population heterogeneity, technical study effects, or interactions between the two; the present design cannot disentangle these sources, nor does the screen itself establish that any individual taxon’s pattern exceeds what a non-independent, overlapping-fold coefficient estimate could show by chance. Standard batch correction methods, designed to remove technical batch effects, would not be expected to resolve directional disagreements to the extent that these differences are driven by population-level biological differences rather than technical artifacts alone — a point that is, if anything, reinforced by Section 4.2’s

finding that label-blind batch correction left cross-cohort AUC essentially unchanged rather than recovering it.

4.5 Implications for Clinical Translation

The results of this study suggest that gut microbiome-based AD classifiers are not ready for cross-population clinical application in their current form. A classifier trained on a Chinese cohort and applied to a Kazakh patient population would perform at or below chance for logistic regression, and only modestly above chance even for LightGBM. These performance levels are clinically inadequate for any diagnostic application.

We propose three prerequisites for meaningful clinical translation of microbiome-based AD diagnostics: (1) **Multi-cohort harmonized training data**: classifiers should be trained on samples from multiple geographic and demographic populations simultaneously, with cohort or population as a stratification variable, rather than training within a single cohort and expecting transfer. (2) **Prospective cross-cohort validation as a publication standard**: any future study reporting a gut microbiome AD classifier should be required to validate performance on at least one independent cohort from a different geographic population before claiming clinical relevance. The within-cohort AUC, no matter how high, is an insufficient evidential standard. (3) **Covariate-adjusted models**: diet, geographic origin, antibiotic history, and other known microbiome-modulating covariates should be incorporated into classification models either as features or confounders, to reduce the influence of population-specific compositional baselines.

The multi-cohort harmonized collection approach has succeeded in other biomarker domains (e.g., the UK Biobank for genomic predictors, the ADNI consortium for neuroimaging). A parallel initiative—perhaps a consortium of AD microbiome studies committed to harmonized collection protocols and prospective cross-cohort sharing—would be the most direct route to a clinically actionable microbiome diagnostic.

4.6 Limitations

Several limitations of this study should be acknowledged explicitly.

Cohort search and selection: The search strategy documented in Section 2.1 and Supplementary Note S1 identified five publicly available cohorts meeting inclusion criteria. The cohort set is not exhaustive of all existing AD microbiome studies, and the inclusion of additional cohorts from non-Asian populations would strengthen the conclusions. The comparability of diagnostic criteria across cohorts (clinical AD diagnosis vs. amyloid-PET positivity) adds biological heterogeneity that cannot be fully disentangled from technical batch effects.

Mixed single-end and paired-end sequencing: Kazakhstan used single-end NovaSeq reads while all other cohorts used paired-end MiSeq 2×300 bp. Single-end reads yield lower taxonomic resolution at equivalent read length due to the absence of overlap-based error correction in DADA2, and different error models were fitted per cohort. This technical asymmetry contributes to between-cohort heterogeneity and cannot be disentangled from biological population differences in this dataset.

Kim/KBASE 2022 excluded from supervised analyses: Per-sample amyloid-PET diagnosis labels are IRB-restricted and not publicly deposited. The Kim 2022 cohort (78 participants; 18 amyloid-positive preclinical AD, 60 CN) therefore contributed only to PERMANOVA variance decomposition.

China overrepresentation: Three of the four labeled cohorts are Chinese. The cross-cohort experiment is therefore primarily a China-internal transfer (Zhuang ↔ Ling ↔ Zhu 2022) plus transfer to/from a single Central Asian cohort. A more geographically balanced design—incorporating North American, European, and African populations—would be needed to generalize the conclusions to the full range of human diversity.

Partial SRA deposit for Zhu 2022: The public archive contains 180 samples representing 90 participants, a subset of the 302 participants analyzed in the published study (PRJNA489760). The deposited subset may not be a random sample of the full study population, potentially introducing selection bias of unknown direction.

Global genus-selection feature universe: The 396-genus feature set (Section 2.2) was selected using $\geq 20\%$ prevalence pooled across all five cohorts simultaneously, meaning the eventual held-out cohort in every LOCO/pairwise experiment contributed prevalence information to the feature universe. A dedicated training-only sensitivity analysis (Section 2.5.1, Section 3.3), run once under the same prespecified `random_state=42` cross-validation partition used throughout this paper, re-derived genus selection independently within every split and found LOCO AUC changed by at most 0.066 — the qualitative generalization-failure conclusion is not an artifact of this design — but the global selection remains a methodological limitation in principle, and we recommend training-only feature selection as the default design for future cross-cohort microbiome classifier studies rather than a post hoc sensitivity check. Repeated cross-validation with multiple random seeds was not performed for this sensitivity analysis, so the 0.066 figure should be read as a single-partition estimate rather than one adjusted for ordinary fold-assignment variability.

Batch correction: two designs, both transductive, neither prospective. As detailed in Section 2.7, both the label-blind (primary) and label-informed (exploratory) batch-correction designs are transductive — both estimate batch-effect parameters using the eventual held-out cohort’s own feature values and cohort membership — and neither should be extrapolated to prospective deployment. The label-informed design additionally uses every sample’s own true diagnosis label during correction and is retained only as an oracle-style sensitivity analysis; its results (including the previously reported severe Kazakhstan collapse under label-informed ComBat-seq) should not be read as evidence about batch correction’s effect on biological signal in general, since they are substantially confounded by the label-informed design itself (Section 4.2).

Cross-sectional design and observational data: All cohorts used cross-sectional sampling. Causal inference about microbiome–AD relationships is not supported by this design. All language throughout this paper refers to associations and predictive performance, not causal effects.

SHAP and coefficient-direction interpretability limitations: For logistic regression, direction is now defined from the fitted model’s coefficient sign (Section 2.8), not from mean signed SHAP; a stable coefficient sign is a statement about what the fitted model consistently learned across cross-validation folds, conditional on regularization (L2, not elastic-net — see below) and collinearity among CLR features, and is not a direct estimate of biological association, its statistical significance, or causal direction. **The $\geq 8/10$ -fold coefficient-sign-stability screen is descriptive, not a formal hypothesis test.** The 10 outer-CV folds within a cohort are fit on substantially overlapping training sets (each omits only 1/10 of that cohort’s samples), so they are not independent statistical replications; we do not apply a multiplicity correction across the candidate taxa screened, and no p-value is assigned to any individual

taxon's stability count. The eight genera meeting this screen and their dietary hypotheses should be treated as empirical patterns worthy of targeted, covariate-adjusted, and independently-replicated follow-up, not as statistically confirmed mechanistic findings. For LightGBM, no coefficient-like global direction exists, and no directional screen of any kind is applied to it in this paper; readers of an earlier draft of this work should disregard any previously reported LightGBM directional-flip count. All LOCO SHAP values (either model) are reported as feature-importance rankings only, without any assigned cross-cohort direction, because the LOCO SHAP background (training cohorts) and evaluation set (held-out cohort) come from systematically different populations, confounding sign with raw cross-cohort compositional differences.

Model regularization mislabeled as “elastic-net” in earlier drafts: The logistic-regression models throughout this paper (`solver="saga"`) were never configured with `penalty="elasticnet"`, and are therefore L2 (ridge)-regularized, not elastic-net, despite being described as “elastic-net” in earlier drafts of this manuscript and despite an `l1_ratio` parameter being present (and silently ignored by scikit-learn) in the estimator configuration. This is a terminology correction, not a re-fit: every AUC, SHAP, and coefficient value reported in this paper was always computed from the L2-regularized model actually fit by the code, and none of those numbers change as a result of this correction. One consequence is that essentially no fitted coefficient is exactly zero in the coefficient-stability analysis (Section 2.8), since L2 regularization does not perform the sparse variable selection that true elastic-net or L1 penalties would.

5. Conclusion

We conducted a systematic cross-cohort generalization evaluation of gut microbiome-based Alzheimer's disease classifiers, spanning five independent cohorts from three countries. Within-cohort classifiers achieved AUC-ROC values of 0.633–0.998, consistent with published reports; however, LOCO cross-cohort AUC values fell to 0.504–0.813, a mean drop of 0.176 AUC points for logistic regression. This generalization-failure pattern was robust to a strict training-only re-derivation of the feature-selection step (LOCO AUC changed by at most 0.066 when held-out-cohort information was fully excluded from genus selection). Batch correction was evaluated under two designs: a label-blind design (no diagnosis information used) left mean LOCO AUC essentially unchanged relative to uncorrected (within 0.007 AUC for both methods and models), and did not recover cross-cohort transferability; an exploratory label-informed design (using each sample's own true diagnosis label, which cannot estimate prospective performance) produced large, cohort-specific degradation that is attributable in substantial part to the label-informed design itself rather than to an intrinsic property of the correction algorithms. PERMANOVA revealed that cohort identity explained approximately 12 times more compositional variance than disease status (marginal R^2 : 0.172 vs. 0.014) — an empirical measure of relative heterogeneity that provides context for, but does not by itself explain, the batch-correction results — and a cohort identity classifier achieved macro-AUC=0.9917, confirming that cohort structure dominates the feature space. Sensitivity analyses confirmed that generalization failure persists across all three prespecified cohort-exclusion configurations. Out-of-fold SHAP analysis identified near-zero but statistically detectable taxon overlap between cohort-specific predictive signatures (mean Jaccard 0.135 for logistic regression and 0.058 for LightGBM, both significant against a null matched to the observed statistic's construction at the primary $N=20$ threshold, though the LightGBM result was not significant at

N=10) and, using a pre-specified descriptive coefficient-sign-stability screen (not a formal hypothesis test), eight genera for logistic regression whose fitted-model coefficient direction is opposite and stable across cohorts; no comparable screen is applied to LightGBM, which has no single coefficient-like global direction. These findings demonstrate that the gut microbiome-based AD classifier literature has a fundamental generalization problem that the two batch-correction methods evaluated here, tested under a fair label-blind design, did not recover. Multi-cohort harmonized prospective studies and mandatory cross-cohort validation are needed before microbiome-based AD biomarkers can achieve clinical utility.

Data Availability Statement

All raw sequencing data analyzed in this study are publicly available from NCBI SRA or ENA under accession numbers PRJNA554111 (Zhuang 2018), PRJNA633959 (Ling 2020), PRJNA489760 (Zhu 2022), PRJNA811324 (Kaiyrlykyzy 2022), and PRJEB50447 (Kim 2022). Analysis code, processed data tables, and figure generation scripts are available at <https://github.com/carhanc/microbiome-ad-generalization>.

Author Contributions

A.C. conceived the study, developed and executed the complete analysis pipeline, interpreted the results, and wrote the manuscript.

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Conflict of Interest

The author declares no conflict of interest.

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